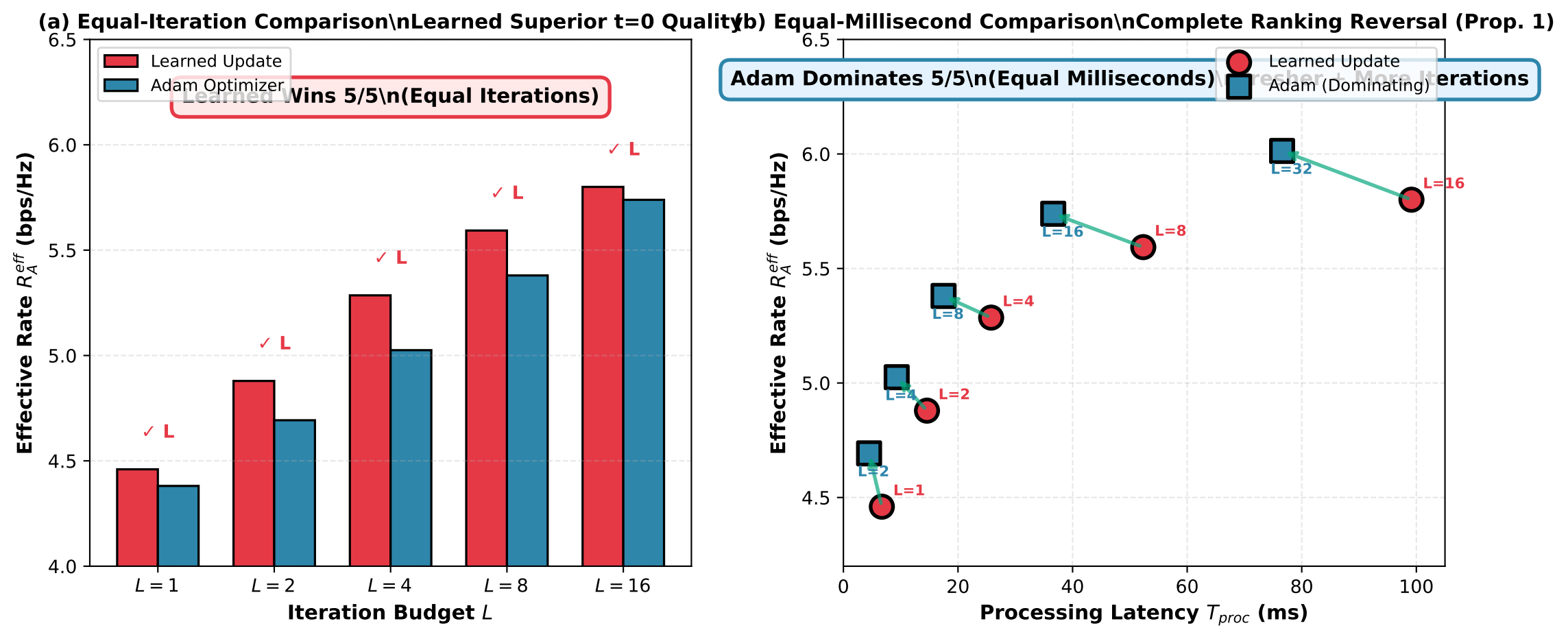
(a) Equal-Iteration Comparison\nLearned Superior t=0 Quality
Learned Update
Adam Optimizer
Learned Wins 5/5\n(Equal Iterations)
✓ L
✓ L
✓ L
✓ L
✓ L
Effective Rate R_A^{eff} (bps/Hz)
6.5
6.0
5.5
5.0
4.5
4.0
$L = 1$
$L = 2$
$L = 4$
$L = 8$
$L = 16$
Iteration Budget L
(b) Equal-Millisecond Comparison\nComplete Ranking Reversal (Prop. 1)
Learned Update
Adam (Dominating)
Adam Dominates 5/5\n(Equal Milliseconds)
Fresher + More Iterations
L=32
L=16
L=16
L=8
L=8
L=4
L=4
L=2
L=2
L=1
Effective Rate R_A^{eff} (bps/Hz)
6.5
6.0
5.5
5.0
4.5
0
20
40
60
80
100
Processing Latency T_{proc} (ms)